Learning to Cooperate in Stag Hunt Gridworlds: An Ablation Study of PPO Design Choices for Two-Agent Coordination

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Abstract

Stag Hunt games capture a core difficulty of multi-agent reinforcement learning: the socially optimal outcome requires every agent to commit to an action that pays off only if the partner commits as well, so independent learners tend to settle on safe but inefficient behaviour. We study how a small set of design choices affects the emergence of cooperation between two PPO agents in the three Gymnasium-Stag-Hunt gridworlds (Hunt, Harvest and Escalation). Starting from a parameter-sharing independent PPO (IPPO) baseline with local critics, we ablate prosocial reward mixing ($c \in \{0.5, 1\}$), the degree of parameter sharing between actors and critics including a centralized critic (MAPPO), handcrafted Manhattan-distance features, and a curriculum on stag mobility. All 29 variants are trained with three seeds, evaluated with a fixed seed-controlled protocol, compared through exact per-agent event counters, and stress-tested under greedy execution and on an unseen larger grid. In Hunt, the baseline never captures the stag, whereas full reward swapping yields 45 captures per episode and a $15\times$ higher team reward, and separate actors with a shared critic recover more than half of that gain without touching the reward. In Escalation, reward mixing, independent actors and MAPPO reach the same plateau of about 95 reciprocal steps per episode, while every variant that keeps a shared actor falls short and collapses under greedy execution. In Harvest, where no dilemma exists, geometric features dominate by shifting the agents from young to mature plants.

1 Introduction

Cooperation between learning agents is fragile when the cooperative action is individually risky. In the classical Stag Hunt [Skyrms, 2004], two hunters each choose between a stag, which can only be caught together and pays a large reward, and a hare, which can be caught alone for a modest reward. Both "stag–stag" and "hare–hare" are Nash equilibria, but the former is payoff dominant while the latter is risk dominant: an agent that explores the stag while its partner forages is punished, and its gradient pushes it towards the safe choice. Peysakhovich and Lerer [2017] showed that in temporally extended Markov versions of this game selfish learners reliably converge to the risk-dominant outcome, and that making agents *prosocial*, i.e. letting them value their partner's reward, restores cooperation.

This project asks a practical question: given a modern on-policy algorithm (PPO, Schulman et al., 2017) and the Gymnasium-Stag-Hunt environments [Nesterov-Rappoport, 2022],[1] which design decisions actually matter for two agents to learn to cooperate? We treat the problem as an ablation study with a fixed training budget and a fixed, seed-controlled evaluation protocol. Concretely, we

[1] https://github.com/giorgiofranceschelli/Gymnasium-Stag-Hunt, built on the Gymnasium API.

contribute (i) a custom two-agent PPO implementation on top of TF-Agents[2] in which parameter sharing, local or joint critics, reward mixing, handcrafted features and an environment curriculum are all selected from a single configuration file; and (ii) a comparison of 29 variants (11 Hunt, 9 Harvest, 9 Escalation), each trained with three seeds and evaluated on 90 episodes against a uniform random floor, plus greedy and larger-grid robustness checks.

2 Background and environments

All three games are played by two agents, A and B, on a 5×5 grid with five actions (left, down, right, up, stay) and episodes of 1000 steps (library defaults). Observations are coordinate vectors (`obs_type=coords`); the environment returns the second observation with the agent slots swapped (`flip_obs`), so each agent always sees itself first and its partner second. Each position is a pair of integer coordinates: the Hunt vector contains the agent, partner, stag and two plant positions (10 values); Harvest the two agents and four plants plus maturity flags (16); Escalation the two agents and the mark (6).

Hunt. The grid contains one stag and two plants. Stepping on a plant gives a forage reward of $+1$ to that agent and the plant respawns. If both agents occupy the stag's cell in the same step, each receives $+5$ and the stag respawns; if only one does, that agent is *mauled* and receives -5. By default the stag moves one cell per step towards the nearest agent. The game thus reproduces the payoff structure of the matrix Stag Hunt: foraging is safe, hunting requires synchrony and lone attempts are costly.

Harvest. Up to four plants are present. Each step a young plant matures with probability 0.1 and a mature one dies with probability 0.1. Harvesting a young plant pays $+1$ to the harvester only; harvesting a mature plant pays $+2$ to *both* agents, whoever reaches it. Cooperation here is not a dilemma: the team obtains a higher joint reward by waiting for plants to mature, although doing so requires patience and coordination.

Escalation. A single *mark* appears on the grid. When both agents stand on it, each receives $+1$, a streak counter s is incremented and the mark moves one random cell. If only one agent is on the mark while $s > 0$, that agent receives $-0.5\, s$ and the streak resets; otherwise rewards are zero. The longer the agents cooperate, the more expensive it becomes for one of them to defect, which is why the game is called escalation [Peysakhovich and Lerer, 2017].

3 Method

3.1 Two-agent PPO

Each agent $i \in \{A, B\}$ observes $o_{i,t}$ and samples action $a_{i,t}$ from $\pi_{\theta_i}(a_{i,t} \mid o_{i,t})$. PPO updates this policy with its clipped objective [Schulman et al., 2017]. At time t, $\pi_{\theta_{i,\text{old}}}$ is the pre-update policy, $\rho_{i,t} = \pi_{\theta_i}(a_{i,t} \mid o_{i,t}) / \pi_{\theta_{i,\text{old}}}(a_{i,t} \mid o_{i,t})$ is the probability ratio, $\hat{A}_{i,t}$ the advantage, and $\hat{R}_{i,t}$ the return. Omitting i below, the minibatch loss is, where $\mathbb{E}_t$ averages minibatch transitions,

$$\begin{aligned} \mathcal{L}(\theta, \phi) = & - \mathbb{E}_t\Big[\min\big(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\, \hat{A}_t\big)\Big] \\ & + \frac{c_v}{2}\, \mathbb{E}_t\Big[(V_\phi(o_t) - \hat{R}_t)^2\Big] - \beta_k\, \mathbb{E}_t\big[\mathcal{H}\big(\pi_\theta(\cdot \mid o_t)\big)\big]\,, \end{aligned} \tag{1}$$

where $\epsilon = 0.2$, $c_v = 1$, and advantages are computed with generalized advantage estimation [Schulman et al., 2016] using $\gamma = 0.99$ and $\lambda = 0.95$, then standardized per rollout (zero mean, unit variance). Following Engstrom et al. [2020], we use orthogonal initialization (gain $\sqrt{2}$ for hidden layers, 0.01 for the policy head, 1 for the value head), global gradient-norm clipping at 0.5, and a learning rate annealed from $3 \cdot 10^{-4}$ to $3 \cdot 10^{-5}$. The entropy coefficient β_k is annealed linearly from 0.15 to 0.01 over 10000 iterations. Actor and critic are separate two-layer MLPs with 128 tanh units.

[2] `https://github.com/tensorflow/agents`

Each iteration collects 128 steps from 6 parallel environments (768 transitions per agent) and performs 4 epochs over minibatches of 64. Over 1000 iterations each seed therefore sees $7.68 \cdot 10^5$ environment steps per agent.

3.2 Ablated design dimensions

Parameter sharing and critic input. The *baseline* (`shared_shared`) uses one actor and one critic for both agents, each conditioned only on the local observation; because observations are flipped, the same network can act for either role. This is independent PPO (IPPO) with parameter sharing, the configuration studied by de Witt et al. [2020] and used as the decentralized reference by Yu et al. [2022]. Three variants relax this sharing at different points: *shared actor, separate critics* keeps a single policy but gives each agent its own value network (a control for critic separation alone), *separate actors, shared critic* does the opposite, and *separate actors, separate critics* is IPPO without any parameter sharing. Orthogonally, the critic can receive the *local* observation o_i or the *joint* observation $[o_A \,\|\, o_B]$ and output one value per agent. The combination of separate actors with a joint critic is the centralized-training/decentralized-execution recipe known as MAPPO [Yu et al., 2022, Lowe et al., 2017]; the combination of a shared actor with a joint critic is reported as *joint critic*. Since o_B is a permutation of o_A, the joint input carries no information that a local critic lacks; these variants therefore act as controls that isolate actor separation from critic form (see Limitations).

Prosocial reward mixing. Following Peysakhovich and Lerer [2017], the rewards fed to the learners are mixed with a coefficient $c \in [0, 1]$:

$$\tilde{r}_A = (1 - c)\, r_A + c\, r_B, \qquad \tilde{r}_B = (1 - c)\, r_B + c\, r_A. \tag{2}$$

With $c = 0$ the agents are selfish, with $c = 0.5$ each optimizes the team mean, and with $c = 1$ the rewards are fully swapped. The environment rewards themselves are untouched; evaluation always reports the original $r_A + r_B$.

Handcrafted geometric features. Coordinates are a poor input for a small MLP because the quantity that matters, relative position, must be learned from differences. The *handcrafted* variant appends to each observation the Manhattan distance $d_{\mathrm{M}}(p, q) = |p_x - q_x| + |p_y - q_y|$ and the signed displacement $(\Delta x, \Delta y)$ between the agent position p and every relevant entity q. In Hunt this adds distances and displacements from both agents to the stag, to the partner and to each plant (padded to a fixed size, 39 inputs in total); in Harvest it adds the partner radar, one slot per plant and the nearest young and nearest mature plant ($d_{\mathrm{M}} = -1$ as a sentinel when absent); in Escalation it adds the partner and the mark radars for both agents (15 inputs).

Stag-mobility curriculum (Hunt only). Catching a pursuing stag with two agents is a hard exploration problem, so we tested a curriculum in the spirit of Bengio et al. [2009] and the environment-shaping curricula surveyed by Narvekar et al. [2020]. The stag's probability of moving in a step is ramped linearly from 0 (a stationary target) to 1 over the first K iterations, with $K = 500$ (*50%*) or $K = 250$ (*25%*); afterwards the default pursuit behaviour is restored, so evaluation always uses the unmodified game. Reward-mixing and stag-mobility curriculum variants retain the baseline `shared_shared` architecture; architecture, critic-input and handcrafted-feature variants use $c = 0$ and no stag curriculum.

4 Experimental protocol

Each of the 29 configurations was trained with seeds $\{42, 43, 44\}$ under the identical hyper-parameters of Section 3. Every trained seed was then evaluated by sampling actions from the final policy on evaluation seeds $\{101, 202, 303\}$ with 10 episodes each, truncated at 100 steps, i.e. 90 episodes per configuration. Unless stated otherwise, all numbers are the mean $\pm$ sample standard deviation *across the three training seeds*, each seed being first averaged over its 30 evaluation episodes; this is the statistic that reflects how reproducible a design choice is. Behavioural metrics are taken from the `info` dictionary of the environments, which we extended to expose exact per-agent event counts (Section 2); for Harvest we additionally count *unique* team harvests so that a mature plant reached by both agents in the same step is counted once. The first row of Table 1 reports what two independent uniform random agents obtain under the same seeds and episode counts; it is the reward floor for all

Table 1: Evaluation team reward per episode (mean $\pm$ std across three training seeds; for the random policy, across the 30 evaluation episodes). Bold marks the best variant per environment. The last two columns are the robustness checks of Section 5.4 as a ratio to the main protocol: Hunt on an unseen 7×7 grid and Escalation under greedy actions.

Variant	Hunt	Harvest	Escalation	Hunt 7×7	Esc. greedy
Uniform random policy (floor)	-171.1 ± 38.0	32.8 ± 14.1	0.1 ± 0.6	–	–
Baseline: IPPO, parameter sharing	28.4 ± 4.7	168.1 ± 2.0	37.3 ± 6.8	0.46	**0.19**
Reward mixing $c = 1.0$	$\mathbf{428.1 \pm 37.6}$	145.9 ± 11.4	181.8 ± 4.8	0.83	0.89
Reward mixing $c = 0.5$	19.7 ± 1.0	160.7 ± 5.0	180.7 ± 1.5	**0.21**	0.75
Shared actor, separate critics	52.1 ± 2.3	223.7 ± 13.6	99.7 ± 51.2	0.60	**0.48**
Separate actors, shared critic	243.1 ± 27.4	206.7 ± 7.5	169.9 ± 17.4	0.64	0.77
Separate actors, separate critics (IPPO)	175.6 ± 82.1	203.8 ± 5.9	$\mathbf{183.3 \pm 2.6}$	0.59	1.00
Joint critic (shared actor)	53.5 ± 0.9	214.0 ± 13.8	53.1 ± 15.3	0.61	**0.17**
Handcrafted features	63.2 ± 0.8	$\mathbf{253.3 \pm 3.1}$	58.4 ± 14.8	0.61	**0.11**
MAPPO (separate actors, joint critic)	210.0 ± 90.5	206.2 ± 9.1	180.2 ± 4.3	0.63	0.98
Stag curriculum, 50% of training	43.1 ± 2.4	–	–	0.59	–
Stag curriculum, 25% of training	42.2 ± 0.9	–	–	0.61	–

results and is strongly negative in Hunt because random agents are frequently mauled. Two robustness checks reuse the same protocol: greedy (argmax) action selection, and evaluation on a 7×7 grid never seen in training.

5 Results

5.1 Hunt: cooperation needs either prosociality or asymmetric policies

Table 1 and Figures 1 and 2 tell a consistent story. The baseline learns to avoid being mauled (from -171 for random play to $+28$), but it does so by foraging: the event counters in Figure 2b show 0.0 stag captures per episode and about 15–20 plants per agent. This is the risk-dominant equilibrium predicted by theory [Skyrms, 2004, Peysakhovich and Lerer, 2017]. Both curriculum variants, the two variants that change only the critic (joint critic, 53.5; separate critics under a shared actor, 52.1) and the handcrafted variant improve foraging efficiency (up to 34 plants per agent) but still never capture the stag; an easier stag early in training did not transfer to hunting once pursuit resumed.

Fully swapping rewards ($c = 1$) changes the picture entirely: it yields 45.0 ± 2.8 stag captures per episode and a team reward of 428, with little variation across seeds (Figure 1, left). A mechanistic reading of Eq. (2) explains why this works where $c = 0.5$ does not. Under full swapping the agent that approaches the stag alone bears no cost (it receives its partner's 0), while the partner receives the -5 and can only turn it into $+5$ by joining the hunt; exploration towards the stag is therefore never discouraged and the partner is actively pulled in. With $c = 0.5$ the lone hunter still internalizes half of the mauling penalty, and in our runs this was enough to keep both agents foraging; the variant actually ends slightly below the baseline (19.7 vs 28.4), the only ablation to do so.

The second route to cooperation does not modify rewards at all: giving the two agents *separate actors* while keeping a shared critic reaches 25.3 captures and a reward of 243, and MAPPO reaches 21.7 captures. We attribute this to symmetry breaking. A shared policy must respond identically to mirrored situations, so when agent A is punished for a solitary attempt the same update discourages B from attempting; with independent parameters, one policy can specialize into a "leader" that commits to the stag while the other learns to follow. The shared-actor/separate-critics control confirms that the actor, not the critic, must be separated: two value networks under one policy stay in the foraging regime. The price of separate actors is variance: with separate critics or a joint critic the seed-to-seed standard deviation is 80–90 reward units; per-seed evaluation rewards range from 85 to 246 with separate critics and from 141 to 313 for MAPPO. The shared local critic appears to be the most stable choice when actors are separate.

5.2 Escalation: many roads to the same plateau, and one dead end

Escalation is the game where the ablations agree most. Reward mixing with either coefficient, separate actors with separate critics and MAPPO all converge to 95 reciprocal mark steps per episode and a team reward of 180–183 with very small seed variance (Figure 3). Because the mark jumps after every joint step, the pair must re-synchronize on a new cell at each success, so the plateau

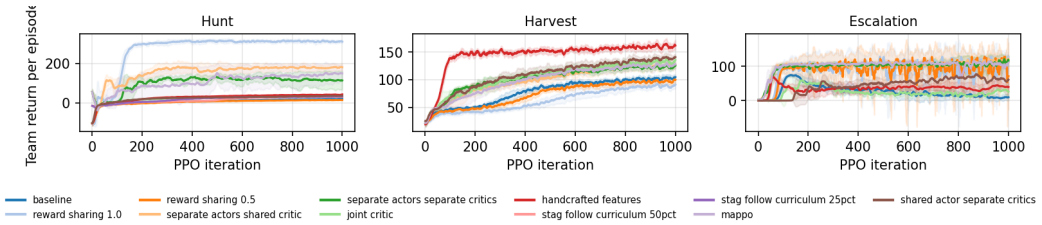
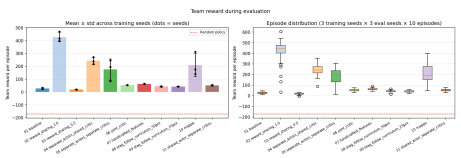


Figure 1: Team return during training (mean $\pm$ std across 3 seeds, moving average of 10 iterations; colours shared across panels). Hunt and Escalation plateau within the budget; several Harvest variants are still improving at iteration 1000.



(a) Evaluation team reward: seed mean $\pm$ std and episode distribution; dashed line: random policy.



(b) Behavioural events per evaluation episode, per agent; captures are counted once per agent.

Figure 2: Hunt. Only the variants that modify rewards or break policy symmetry leave the foraging regime.

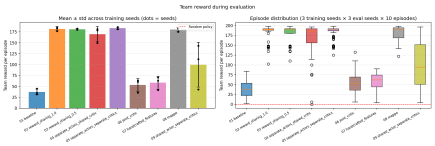
reflects a learned pursuit routine rather than reward saturation. Streak-break punishments are rare in all cooperative variants (≤ 0.2 per episode) but large (6–9 reward units), i.e. the few breaks happen after long streaks, when a single mistimed step is expensive. The baseline achieves only 18.7 reciprocal steps per episode; its near-zero streak-break punishment indicates that it rarely builds long cooperative streaks, although the counts alone do not establish whether it learned to avoid the mark when alone.

The clear negative result is the *joint critic with a shared actor* (53), echoed by the handcrafted variant (58). Both keep the baseline's shared policy and change only its inputs, leaving untouched the symmetric credit assignment that, as in Hunt, makes the shared update conservative. Giving the shared actor two separate critics helps only erratically (99.7 ± 51.2; per-seed rewards 43, 113, 143, none at the plateau). Training is noisier here than elsewhere (Figure 1, right): separate critics show large seed variance in training (80 ± 62 final return) despite a stable evaluation (183 ± 3). MAPPO uses the same joint critic as the failing variant but with separate actors and succeeds, which isolates actor separation, not the choice of critic, as the ingredient that matters in this game.

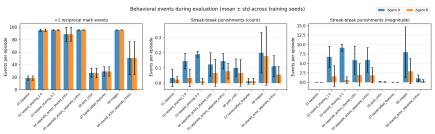
5.3 Harvest: a coordination task without a dilemma

Every Harvest variant is far above the random floor and even the baseline reaches 168 (Figure 4a). Most Harvest curves are still rising at iteration 1000 (Figure 1, centre), so the ranking is at a fixed budget rather than at convergence. The ranking is driven by representation and value estimation, not prosociality: the handcrafted variant is best by a wide margin (253.3 ± 3.1), followed by the two variants that change only the critic, separate critics under a shared actor (224) and a joint critic (214), and by the three separate-actor variants (204–207). Reward mixing is mildly harmful (146 and 161), which is expected since mature harvests are already shared by the environment and mixing only adds noise to the individual young-plant signal.

The unique-team-harvest counters in Figure 4b explain the handcrafted result. Compared with the baseline the geometric agents harvest fewer young plants (18 vs 28 unique per episode) and far more mature ones (41 vs 29). Having the nearest mature and nearest young plant as explicit inputs lets the policy weigh $+2$-to-both against $+1$-to-self and wait for maturity, which the coordinate-only baseline never discovers within the budget. Per-agent evaluation rewards differ by less than two units in every variant, so there is no evidence of free-riding.

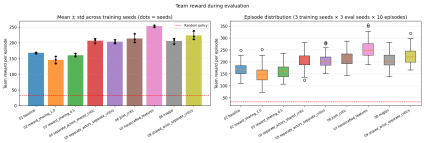


(a) Evaluation team reward (seed mean ± std; episode distribution).

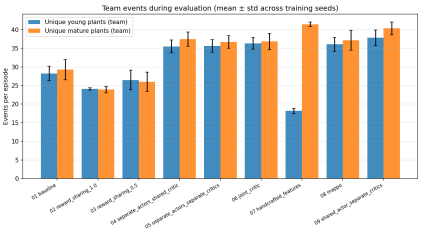


(b) Behavioural events per evaluation episode.

Figure 3: Escalation. Reward mixing, independent actors and MAPPO reach the same plateau; variants that keep a shared actor do not.



(a) Evaluation team reward (seed mean ± std; episode distribution).



(b) Unique young and mature plants harvested by the team.

Figure 4: Harvest. Geometric features shift the agents from young to mature plants and give the best team reward.

5.4 Robustness: greedy execution and a larger grid

Greedy (argmax) execution is a fragility test. In Escalation every shared-actor variant collapses under it (Table 1, last column: baseline $37 \to 7$, joint critic $53 \to 9$, handcrafted $58 \to 7$), reward mixing loses 11–25%, while separate actors keep their plateau (0.98–1.00): shared policies rely on stochastic re-synchronization, independent actors on a deterministic pursuit routine. In Hunt and Harvest argmax never hurts (ratios 0.99–1.40). On an unseen 7×7 grid, Escalation transfers unchanged (0.9–1.1), Harvest drops uniformly to about 0.6 (fewer plants per cell), and Hunt degrades most: the baseline halves and $c = 0.5$ loses 79%, while full reward swapping retains 83% and remains the only variant that still hunts. The ranking of the main study is preserved everywhere.

6 Discussion

Three observations cut across the games. First, the interventions that help change *how credit flows between the agents*: prosocial reward mixing does so explicitly, separate actors implicitly by letting the two policies receive different gradients. Changing only the critic (its input or its sharing) or the observation leaves the baseline's shared, symmetric policy update untouched and solves neither dilemma, although it is the best choice in Harvest, where no dilemma exists. Second, reward mixing is environment-dependent: $c = 0.5$ works in Escalation but not Hunt, where only full swapping ($c = 1$) yields a large gain. Third, stability and performance trade off: shared-critic variants have the smallest variance among the cooperative designs, while separate or joint critics occasionally leave a seed near the safe equilibrium.

Limitations. Results use three training seeds per variant on a 5×5 grid with default rewards; more seeds would sharpen the high-variance comparisons. Because the games are fully observable, the joint critic never holds information the actor lacks, so the MAPPO comparison does not test the informational advantage behind centralized critics. The curriculum used one fixed schedule, entropy was not tuned per variant, and agents were always evaluated with their training partner.

7 Conclusion

Under a fixed, seed-controlled protocol, prosocial reward swapping is the most effective, stable and transferable solution in Hunt; reward mixing and separate actors suffice in Escalation, where only separate actors survive greedy execution; geometric features win in Harvest. At equal information, changing only the critic and the tested curriculum solve neither dilemma.

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